VeloCraft

Race Control

Live gamified scoreboard for SAP S/4HANA Public Cloud training

How the dashboard works and how to run a session

What it is

VeloCraft Race Control is a live, cycling-race-themed scoreboard for a hands-on SAP S/4HANA Public Cloud training session. Trainees work in teams through SAP tasks dressed up as race "stages". They log their answers and climb the "general classification" on a big projector leaderboard. The current leader wears the yellow jersey.

It is a single web application. One server powers three pages, and every screen updates live the instant anything changes — no refreshing, no page reloads.

The three views

- Team view (/) — one per table of trainees. Teams sign in with a name and a 4-character code, read their brief, solve stages, and request hints.
- Leaderboard (/board) — the projector screen. Shows the ranked "general classification" and a live "Race Radio" feed of solves. Read-only; just put it on screen.
- Facilitator console (/host) — your control panel, protected by an admin password. Runs the clock, manages stages and teams, unlocks hints, and records manual answers.

How a session runs

- Open /host and log in with the admin password.
- Set the race clock length (default 60 minutes) and leave it idle.
- Build or confirm your stages, and add the accepted answer(s) to each one.
- Hand each table their team name and 4-character code. Put /board on the projector.
- Press Start. Questions unlock on every team screen instantly, and the clock counts down.
- Teams complete tasks in the real SAP system, find the answer, and submit it. The board and radio update live.
- Unlock hints as requests come in. Use +5 min for more time, or End now to stop early.
- When the clock reaches zero the race ends automatically and no more points are scored.

Stages, drafts and answers

A stage is one SAP task. A stage stays a hidden DRAFT — invisible to teams — until it has at least one accepted answer. This lets you pre-load all your stages and reveal each one simply by adding its answer. You can list several acceptable answers separated by a vertical bar, for example: Century Cycles|Century-Cycles

Answers are matched forgivingly. The server lowercases the text, strips accents, and removes spaces and punctuation before comparing — so " century-CYCLES " still matches "Century Cycles". Teams never receive the answers; their browser only ever learns whether a guess was right or wrong.

Scoring

Base points — Easy / Medium / Hard stage	**5 / 10 / 15**
Solved without using a hint	**+5**
First team to solve a given stage	**+3**
First team to solve every live stage	**+10**

Each team gets 3 hint tokens by default. Requesting a hint spends a token, but the hint text only appears once the facilitator unlocks it. Ties are broken by number of stages solved, then by who reached their last solve earliest.

How it updates live

The server holds an open connection to every browser (Server-Sent Events). It sends a clock "tick" every second to keep all timers in sync and to auto-end the race, and an "update" message whenever something changes — a solve, a new stage, a hint unlock. Each page listens and redraws itself automatically. The projector also quietly re-checks every five seconds as a safety net.

Where the data lives

All state is stored in a single SQLite database file at data/race.db, created automatically on first run and seeded with a demo stage and four sample teams. Accepted answers live in the "answers" column of the stages (missions) table. Editing a stage in the facilitator console writes straight to this database — you never edit code to change questions or answers during a session.

Because everything is one file, scores survive a server restart. On a cloud host, attach a persistent disk so the file is not wiped on redeploy.

Security notes

- The admin password lives only in a .env file and is never committed to source control.
- Teams authenticate with their 4-character code; the facilitator with the admin password.
- Answers are checked on the server and are never sent to team browsers, so they cannot be found by viewing page source.
- Keep the facilitator console off the projector during the session.